

Helium gas or liquid nitrogen for the first thermal shield of a 4 K cryostat: conjugate heat-transfer budget, coolant operating maps and work ledger

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Abstract

The first thermal shield of a 4 K cryostat intercepts most of the heat arriving from the 300 K environment. It can be held near 50 K by circulated helium gas, as in fusion and accelerator cryostats, or at 77 K by liquid nitrogen, as in magnet cryostats. This paper separates that choice into three questions and answers each with a dedicated model. (1) The passive penalty of the warmer shield is computed with a three-dimensional multi-region conjugate heat-transfer model of a representative cryostat section that resolves view-factor radiation and the conduction of a 65-line coaxial harness; the view-factor matrix is shown to require row normalisation in a low-emissivity enclosure, after which the model reproduces the closed-form loads to 4 %. (2) The two coolants are compared in a channelled cold plate at a matched 9.4 W duty over Reynolds numbers 500–10 000, laminar to turbulent, and helium pressures of 1–18 bar, with properties from the National Institute of Standards and Technology (NIST) reference equations of state, three-grid verification for both fluids and energy closure below 0.2 %. (3) The routes are priced in room-temperature power with ideal Carnot multipliers and with published hardware efficiencies. The warmer shield costs nothing at the first stage, 8.3 against 8.4 W; its penalty is 106 mW at the 4 K stage, 86 % of it harness conduction, so it is a property of the wiring rather than of the cryogen. In the cold plate nitrogen gives five times helium’s heat-transfer coefficient in laminar flow and six times in turbulent flow, and holds the plate within 4 K of its inlet where helium at 1 bar floats 21 K above and exceeds 77 K below a

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Reynolds number of about 1400; at 18 bar helium's pumping power falls 8.5 times below nitrogen's with heat transfer unchanged, so the ranking depends on the basis and the operating point. The ideal ledger favours nitrogen by 21 % for a bare shield and helium by 8 % once multi-layer insulation removes most of the first-stage load; with hardware efficiencies helium is cheaper by 35–50 % in both cases, and nitrogen becomes competitive only if the harness conduction to 4 K is cut by about a factor of three. Operating maps, uncertainty bands and the complete case set are provided.

Keywords: cryostat, thermal shield, conjugate heat transfer, liquid nitrogen, helium gas, pulse-tube cryocooler, exergy, OpenFOAM

Nomenclature

Roman

A	area [m ²]
c_p	specific heat [J kg ⁻¹ K ⁻¹]
D_h	hydraulic diameter [m]
f	Darcy friction factor [-]
F_{ij}	view factor from face i to face j [-]
h	heat-transfer coefficient [W m ⁻² K ⁻¹]
J	radiosity [W m ⁻²]
k	thermal conductivity [W m ⁻¹ K ⁻¹]
$\dot{m}$	mass flow rate [kg s ⁻¹]
Mo	Mouromtseff number [-]
N	number of MLI layers [-]
NTU	number of transfer units [-]
Nu	Nusselt number hD_h/k [-]
p	pressure [Pa]
Pr	Prandtl number [-]
$\dot{Q}$	heat load [W]
q	heat flux [W m ⁻²]
Re	Reynolds number $\rho U D_h/\mu$ [-]
$\dot{S}_{\text{gen}}$	entropy generation rate [W K ⁻¹]
T	temperature [K]
T_0	heat-rejection temperature, 300 K
U	mean velocity [m s ⁻¹]
$\dot{W}$	wall power [W]
$\dot{W}_p$	pumping power [W]
y^+	wall coordinate [-]

Greek

β	thermal expansion coefficient [K ⁻¹]
ε	emissivity; HX effectiveness [-]
η_i	stage efficiency, fraction of Carnot [-]
κ_{eff}	effective harness conductivity [W m ⁻¹ K ⁻¹]
μ	dynamic viscosity [Pa s]
ρ	density [kg m ⁻³]
σ	Stefan–Boltzmann constant

Subscripts 0 ambient; 1 first stage; 2 second (4 K) stage; b bulk; c conduction; r radiation; sat saturation; w wall.

Acronyms

CFD computational fluid dynamics; CHT conjugate heat transfer; DTT Divertor Tokamak Test facility; GCI grid convergence index; ITER International Thermonuclear Experimental Reactor; LHC Large Hadron Collider; LHD Large Helical Device; MLI multi-layer insulation; NIST National Institute of Standards and Technology; NTU number of transfer units; OVC outer vacuum can; PTFE polytetrafluoroethylene; PTR pulse-tube refrigerator; SIMPLE semi-implicit method for pressure-linked equations; SST shear-stress transport; STL stereolithography (surface file format).

1. Introduction

A cryostat that holds a payload at 4 K places at least one thermal shield between the payload and the 300 K vacuum vessel. The shield intercepts the radiation from the vessel and the conduction along supports and wiring, and it does so at a temperature where refrigeration is one to two orders of magnitude cheaper than at 4 K [1, 2, 3]. Two ways of cooling this first shield are in wide use. Large superconducting systems circulate pressurised helium gas through cooling tubes on the shield: the thermal shield of the International Thermonuclear Experimental Reactor (ITER) runs on helium at 80 K and 1.8 MPa [4], the shield of the Divertor Tokamak Test facility (DTT) on the same loop conditions [5], the shields of the Large Helical Device (LHD) on gaseous helium in parallel paths [6], and the shields of the Large Hadron Collider (LHC) on helium at 50–75 K [7, 8]. Magnet and detector cryostats, and the older generation of helium cryostats, use a liquid-nitrogen bath or loop at 77 K [9, 10]; Li et al. made the practical case for it three decades ago, that a boiling cryogen holds the shield temperature constant and carries a stored reserve, whereas a helium-gas shield warms as soon as its refrigerator stops [11]. Compact laboratory cryostats, including the dry dilution refrigerators that host superconducting quantum processors, use the first stage of a helium pulse-tube refrigerator (PTR) at 40–50 K instead [12, 13, 14]. As these laboratory systems grow toward the multi-metre cryostats now being built for quantum platforms [15, 16], they inherit the shield-cooling question that fusion and accelerator cryostats answered

with circulated fluids, and the cryogenic plant becomes a dominant part of the system's energy cost [17, 18, 19].

The published treatments of this choice fall into four strands that do not meet. First, shield design temperatures are optimised with lumped static budgets and refrigeration-power minimisation: Lagrange-multiplier optimisation of nitrogen, neon and helium-gas intercepts on supports and in insulation [11], multi-stage shields for transmission-line cryostats [20], fusion thermal shields [21, 5], and single-shield helium cryostats brought to zero boil-off by load minimisation [22]. These are one-dimensional: they return a stage total, not where the load lands. Second, wiring loads are measured and modelled per cable [14, 23, 24], again as conduction integrals that assign the harness a single resistance and leave it uncoupled from the radiation field. Third, surface-to-surface radiation in computational fluid dynamics (CFD) codes has been validated on enclosure benchmarks and applied to fusion vacuum vessels [25], but not coupled to temperature-dependent harness conduction in a cryostat shield, and the behaviour of agglomerated view-factor matrices in such enclosures has not been examined. Fourth, cryogenic heat exchangers are optimised by entropy-generation minimisation [26, 27, 28] and liquid-nitrogen cooled components are now modelled with combined analytical and CFD frameworks [29], yet helium gas and liquid nitrogen have not been ranked against each other as forced-convection first-shield coolants on a common duty across flow regimes and pressures, with heat-transfer, hydraulic, effectiveness and second-law metrics reported together and the single-phase margin of the nitrogen verified. Finally, the cryocooler and liquefier efficiencies that separate an ideal Carnot ledger from a real one are well documented [30, 8, 31], but the two ledgers have not been reconciled for this choice.

The gap is therefore not a missing number but a missing junction. What is absent is a treatment that (i) resolves shield radiation and harness conduction together in three dimensions on one geometry, (ii) compares the two coolants like for like in the cold plate that geometry implies, and (iii) prices both routes with ideal and with hardware efficiencies, so that the conditions under which each route wins can be stated rather than assumed.

This paper therefore asks three questions about a representative 4 K cryostat section with a 65-line coaxial harness (Figs. 1 and 2):

Q1 (passive penalty). What does raising the first shield from 50 K to 77 K cost in heat load at the first stage and at the 4 K stage, and which

61 mechanism carries the cost?

62 **Q2 (coolant performance).** If the first-stage duty is removed by a coolant
 63 circulated through a channelled cold plate, how do helium gas and
 64 liquid nitrogen compare across flow regimes, pressures and bases of
 65 comparison, and where is each fluid’s best operating point?

66 **Q3 (work ledger).** What do the two routes cost in room-temperature power,
 67 with ideal Carnot multipliers and with the efficiencies of real hardware?

68 Q1 is answered with a three-dimensional multi-region conjugate heat-transfer
 69 (CHT) model that resolves view-factor radiation and temperature-dependent
 70 harness conduction. Its stage-integrated loads are verified against closed-
 71 form budgets; its contribution is the spatial distribution of the load and the
 72 split between harness families. Q2 is answered with a grid-converged cold-
 73 plate model run for both fluids over $Re = 500\text{--}10^4$ and helium pressures of
 74 1–18 bar, with properties from the NIST reference equations of state [32]. Q3
 75 combines the stage loads of Q1 with Carnot multipliers and with published
 76 cryocooler and liquid-nitrogen plant efficiencies [30, 33, 31]. The dilution
 77 unit or any payload below 4 K is outside the scope; the results apply to any
 78 4 K cryostat with a wired first shield.

79 2. Models and methods

80 2.1. Governing equations

81 Both models solve steady conservation equations with OpenFOAM v2412 [34].
 82 In the enclosure model the inter-stage gaps are vacuum, so each gap region
 83 carries only the radiative exchange and each solid region carries conduction
 84 with a temperature-dependent conductivity,

$$\nabla \cdot [\kappa_{\text{eff}}(T) \nabla T] = 0 \quad \text{in the harness bundles,} \quad (1)$$

85 coupled at the bundle surfaces to the radiative flux of the enclosing gap.
 86 Radiation is solved as a grey-diffuse radiosity system on the agglomerated
 87 boundary faces,

$$J_i = \varepsilon_i \sigma T_i^4 + (1 - \varepsilon_i) \sum_j F_{ij} J_j, \quad q_{r,i} = J_i - \sum_j F_{ij} J_j, \quad (2)$$

88 where F_{ij} are geometric view factors computed by ray tracing between ag-
 89 glomerated faces (**viewFactor** model). For two parallel grey surfaces Eq. (2)

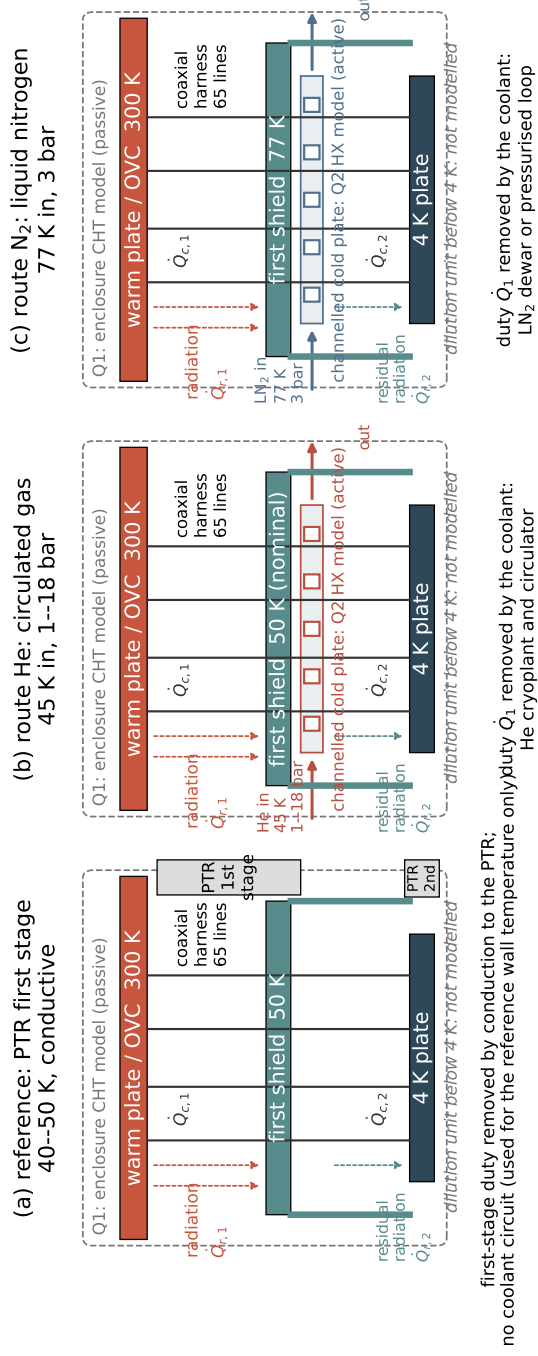


Figure 2: The three first-shield cooling routes and the two models. (a) Reference architecture with a conductively coupled PTR first stage; (b) circulated helium gas through a channelled cold plate; (c) liquid nitrogen through the same cold plate. The dashed box is the domain of the passive enclosure CHT model (Q1); the dotted box is the domain of the active cold-plate model (Q2). The loads $\dot{Q}_1 = \dot{Q}_{r,1} + \dot{Q}_{c,1}$ and $\dot{Q}_2 = \dot{Q}_{r,2} + \dot{Q}_{c,2}$ enter the work ledger ($\dot{Q}_3$), $\dot{W} = \sum_i \dot{Q}_i (T_0 - T_i) / (T_i \eta_i)$, with $\eta_i = 1$ (ideal) or the measured fraction of Carnot (real).

90 reduces to the closed form $q_r = \sigma(T_{\text{hot}}^4 - T_{\text{cold}}^4)/(1/\varepsilon_{\text{hot}} + 1/\varepsilon_{\text{cold}} - 1)$ [35],
 91 which is used as the verification anchor.

92 The cold-plate model solves the steady Reynolds-averaged (or, in the
 93 laminar range, the plain) continuity, momentum and energy equations for a
 94 single-phase coolant,

$$\nabla \cdot (\rho \mathbf{U}) = 0, \quad (3)$$

$$\nabla \cdot (\rho \mathbf{U} \mathbf{U}) = -\nabla p + \nabla \cdot [(\mu + \mu_t)(\nabla \mathbf{U} + \nabla \mathbf{U}^T)] + \rho \mathbf{g}, \quad (4)$$

$$\nabla \cdot (\rho \mathbf{U} h) = \nabla \cdot \left[\left(\frac{k}{c_p} + \frac{\mu_t}{Pr_t} \right) \nabla h \right], \quad (5)$$

95 with $\mu_t = 0$ for laminar runs and μ_t from the k - ω shear-stress transport
 96 (SST) model [36] for $Re \geq 5000$ ($Pr_t = 0.85$). The sweep is run as pure forced
 97 convection ($\mathbf{g} = 0$) so that the ranking does not depend on the orientation
 98 of the plate; the buoyancy parameter Gr/Re^2 is reported for every run, and
 99 three runs with gravity acting along the heated wall quantify the mixed-
 100 convection sensitivity (Section 4). Helium is treated as an ideal gas with
 101 variable density, $\rho = pM/(R_u T)$, because its density changes by 5–26 %
 102 between inlet and outlet and by up to a factor of 2 between bulk and heated
 103 wall; liquid nitrogen is treated with the Boussinesq approximation, $\rho = \rho_0[1 -$
 104 $\beta(T - T_0)]$, valid because $\beta\Delta T < 0.03$ in every run.

105 2.2. Enclosure CHT model (Q1)

106 The domain is a $0.718 \times 0.718 \times 0.335$ m section of a cryostat between
 107 the warm plate and the first-stage cold plate (Fig. 4; Table 1). Two vacuum
 108 gap regions (outer: outer vacuum can (OVC) to shield; inner: warm plate to
 109 cold plate and shield) and four solid cylinders of 20 mm diameter spanning
 110 the 0.335 m height are meshed with a $30 \times 30 \times 14$ background grid refined
 111 to level 2 around the cylinders ($0.2\text{--}0.4 \times 10^6$ cells in total). Each cylinder
 112 lumps one functional family of a cabled 65-line control harness [14]: 25 drive,
 113 25 flux, 10 readout and 5 pump lines, all of the UT-085 stainless-steel/PTFE
 114 coaxial construction. The family is represented by an effective conductivity

$$\kappa_{\text{eff}}(T) = \frac{n_k}{A_{\text{model}}} \sum_i A_i k_i(T), \quad (6)$$

115 where n_k is the number of lines in the family, A_i the cross-section of cable
 116 component i (outer conductor 1.585 mm^2 and centre conductor 0.205 mm^2)

Table 1: Enclosure model: geometry, boundary conditions and numerical settings.

Domain	$0.718 \times 0.718 \times 0.335$ m; OVC and shield as snapped stereolithography (STL) surfaces; four coax cylinders $\varnothing 20$ mm at radius 0.15 m
Regions	<code>domain0</code> (OVC–shield gap), <code>domain1</code> (warm plate–cold plate/shield gap), <code>coax.L1..L4</code> (solids)
Gas in gaps	vacuum: $p = 10^{-3}$ Pa, $g = 0$, $\mu = 10^{-11}$ Pa s (no convection, negligible gas conduction; free-molecular bound < 1 % of the stage load, Supplementary S1)
Temperatures	fixed: warm plate and OVC at T_{hot} ; shield and cold plate at T_{cold} ; cases (300, 50), (300, 77), (50, 4), (77, 4) K
Emissivity (baseline)	warm surfaces 0.04 (polished Al); shield and cold plate 0.02 (polished Cu-plated steel); harness 0.02
Emissivity (variants)	± 50 % (0.02/0.01 and 0.06/0.03); multi-layer-insulation (MLI) equivalent 0.003 on all first-stage surfaces
Radiation	<code>viewFactor</code> , grey diffuse, one band, solved every 10 iterations; agglomeration 250 faces per patch at the coarsest level (verification at 400 and 800)
Harness contact	bundles fixed at T_{hot} at the warm cap and T_{cold} at the cold cap; no contact resistance; lateral surfaces exchange by radiation only
Solver	<code>chtMultiRegionSimpleFoam</code> , semi-implicit method for pressure-linked equations (SIMPLE), residuals $< 10^{-7}$, 2900 iterations from a uniform initial field

117 of 304 stainless steel, polytetrafluoroethylene (PTFE) dielectric 2.001 mm^2),
118 $k_i(T)$ the NIST cryogenic conductivity fits, and A_{model} the effective conduc-
119 tion area of the modelled cylinder, calibrated once per geometry so that
120 the resolved conduction reproduces the harness integral $n_k L^{-1} \sum_i A_i \int k_i dT$
121 (Supplementary Table S4 lists the calibrated areas and the fifth-order poly-
122 nomial coefficients of κ_{eff} for every bundle and stage span; fit error below
123 0.5 %; Fig. 3).

124 *Verification and uncertainty.* Stage-integrated radiation is compared with
125 the parallel-plate closed form and harness conduction with the analytic in-
126 tegral; both agree within the tolerances reported in Section 3. Because the
127 view-factor matrix is built on agglomerated faces, the radiative load was re-
128 computed at 250, 400 and 800 coarse faces per patch and its closure $\sum_j F_{ij}$

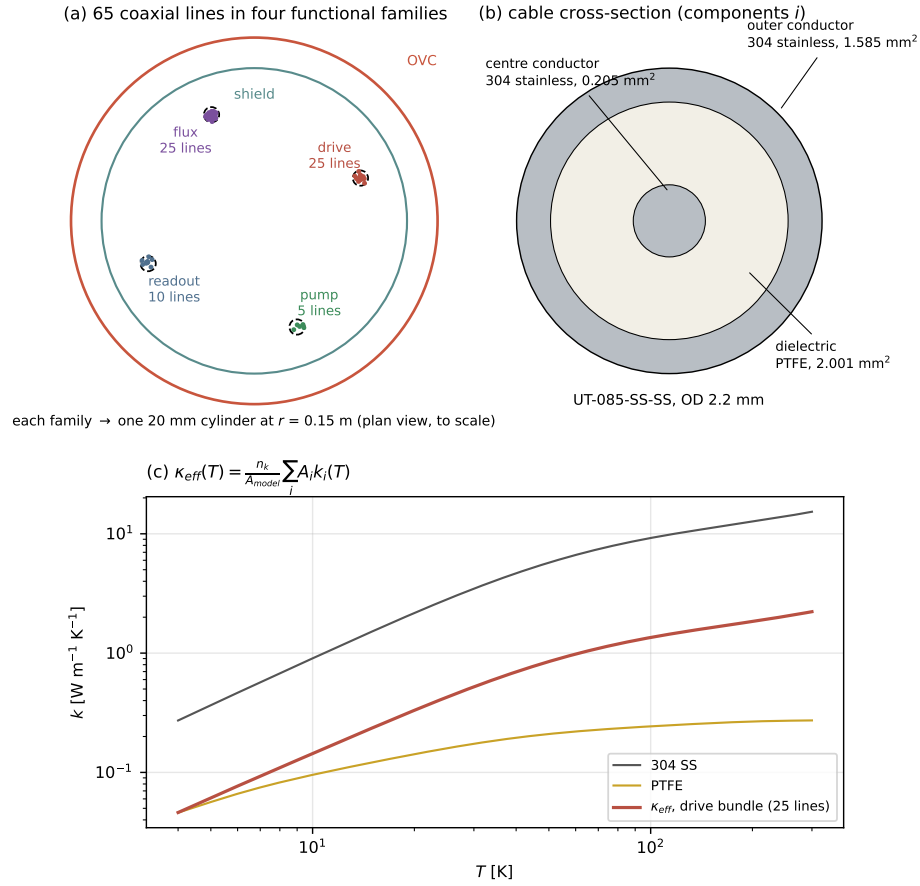


Figure 3: Harness representation: (a) the 65 coaxial lines in four functional families, each lumped into one 20 mm cylinder; (b) the UT-085 cable cross-section with the component areas of Eq. (6); (c) the component conductivities and the resulting effective conductivity of the drive bundle.

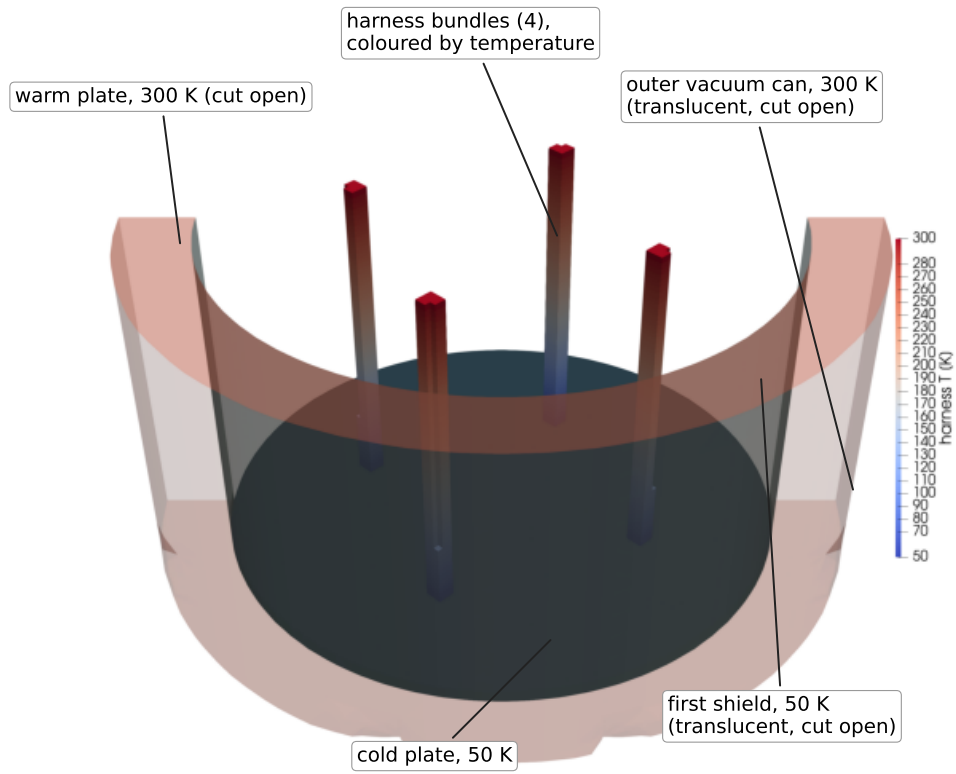


Figure 4: Cut-away of the enclosure CHT domain for the 300 $\rightarrow$ 50 K case. The warm plate, outer vacuum can and first shield are cut open on the near side; the four harness bundles, coloured by their computed temperature, span the gap from the warm plate to the cold plate. The vacuum gaps between the surfaces carry radiation only.

monitored (Supplementary Table S5). The emissivity of polished cryogenic surfaces is uncertain by roughly a factor of 1.5 [37]; the $\pm 50\%$ runs give the uncertainty band carried through the ledger. An MLI-equivalent variant with $\varepsilon = 0.003$ on the first-stage surfaces represents a 10–20 layer blanket [38, 1].

What the three-dimensional model adds.. For a parallel-plate enclosure with fixed wall temperatures, the stage-integrated radiative and conductive loads are analytic, and the CHT model must reproduce them; that agreement is the model’s verification, not its result. The model contributes what the closed forms cannot: the distribution of radiative flux over the shield (which sets local shield temperatures once a real, finitely conductive shield is considered), the partition of conduction among harness families, the coupling between the harness and the surrounding radiation field, and, through the same geometry, the boundary conditions passed to the cold-plate model.

2.3. Cold-plate heat-exchanger model (Q2)

Basis of comparison.. The two fluids differ in phase, temperature, density by a factor of 800 and thermal conductivity by a factor of 3.1, so no single common basis is neutral. Three bases are therefore reported: matched Reynolds number (same flow regime), matched mass flow, and matched pumping power. All runs remove the same duty of 9.4 W, the conservative first-stage load of Section 3. Helium enters at 45 K, at 1, 5 or 18 bar (the ITER shield loop runs at 18 bar [4]); nitrogen enters as a liquid at 77 K in a 3 bar loop, for which $T_{\text{sat}} = 87.9$ K, and the saturation margin $T_{\text{sat}} - T_{\text{w,max}}$ is reported for every run to confirm single-phase operation. Table 2 lists the operating quantities and Fig. 5 the geometry.

Property correction.. An earlier version of this comparison used helium transport properties that correspond to about 15 K ($\mu = 2.7 \mu\text{Pa s}$, $k = 0.019 \text{ W m}^{-1}\text{K}^{-1}$), which understated the helium heat-transfer coefficient by a factor of about 2.5. All helium results here use the 50 K values of Table 2.

Metrics.. The heat-transfer coefficient is $h = \dot{Q}/[A_{\text{heated}}(T_{\text{w}} - T_{\text{b}})]$ with T_{w} the area-averaged heated-wall temperature and T_{b} the mean of the inlet and the mass-flux-weighted outlet temperature; $Nu = hD_{\text{h}}/k$; the Darcy friction factor $f = 2\Delta p D_{\text{h}}/(\rho U^2 L)$; pumping power $\dot{W}_{\text{p}} = \dot{m}\Delta p/\rho$; $NTU = hA/(\dot{m}c_p)$ and $\varepsilon = 1 - e^{-NTU}$ [39]; the Mouromtseff number $Mo = k^{0.6}\rho^{0.8}c_p^{0.4}\mu^{-0.4}$ [40]; and the exergy destruction $T_0\dot{S}_{\text{gen}}$ from volume integrals of the thermal and viscous entropy sources [28]. Laminar results are checked against the fully

Table 2: Cold-plate model: geometry, coolant states, properties and numerics. Properties from the NIST reference equations of state via CoolProp [32], evaluated at the inlet state and held constant except for the helium density (ideal gas) and the nitrogen buoyancy term.

	Helium gas	Liquid nitrogen
Geometry	five parallel square ducts, $D_h = 10$ mm, $L = 0.30$ m, one wall heated, $A_{\text{heated}} = 0.015 \text{ m}^2$, $A_{\text{cs}} = 5 \times 10^{-4} \text{ m}^2$	
Duty	9.4 W uniform flux on the heated wall, $q_w = 627 \text{ W m}^{-2}$	
Inlet T , p	45 K; 1, 5, 18 bar	77 K; 3 bar ($T_{\text{sat}} = 87.9 \text{ K}$)
ρ [kg m^{-3}]	0.961 (1 bar), 4.80 (5 bar), 17.3 (18 bar) at 50 K	808
c_p [$\text{J kg}^{-1} \text{K}^{-1}$]	5201	2038
μ [$\mu \text{Pa s}$]	6.36	163.5
k [$\text{W m}^{-1} \text{K}^{-1}$]	0.0467	0.1457
Pr	0.71	2.29
β [K^{-1}]	$1/T$ (ideal gas)	5.6×10^{-3}
Re	500, 1000, 1500, 2300 (laminar); 5000, 10 000 (k - ω SST)	
Gravity	off (forced convection) for the sweep; on, parallel to the heated wall, for three sensitivity runs	
$\dot{m}$ at $Re = 2300$ [kg s^{-1}]	7.31×10^{-4}	1.88×10^{-2}
U at $Re = 2300$ [m s^{-1}]	1.52 (1 bar), 0.084 (18 bar)	0.047
Solver	buoyantSimpleFoam, heRhoThermo/perfect gas	buoyantBoussinesqSimpleFoam
Mesh	180 $\times$ 18 $\times$ 18 cells per duct, two-sided wall grading (ratio 6), 2.92×10^5 cells; grid convergence index (GCI) study on 86k/292k/984k cells for both fluids	
Convergence	residuals $< 10^{-6}$ (He) and $< 10^{-7}$ (LN ₂); energy balance closure reported per run	

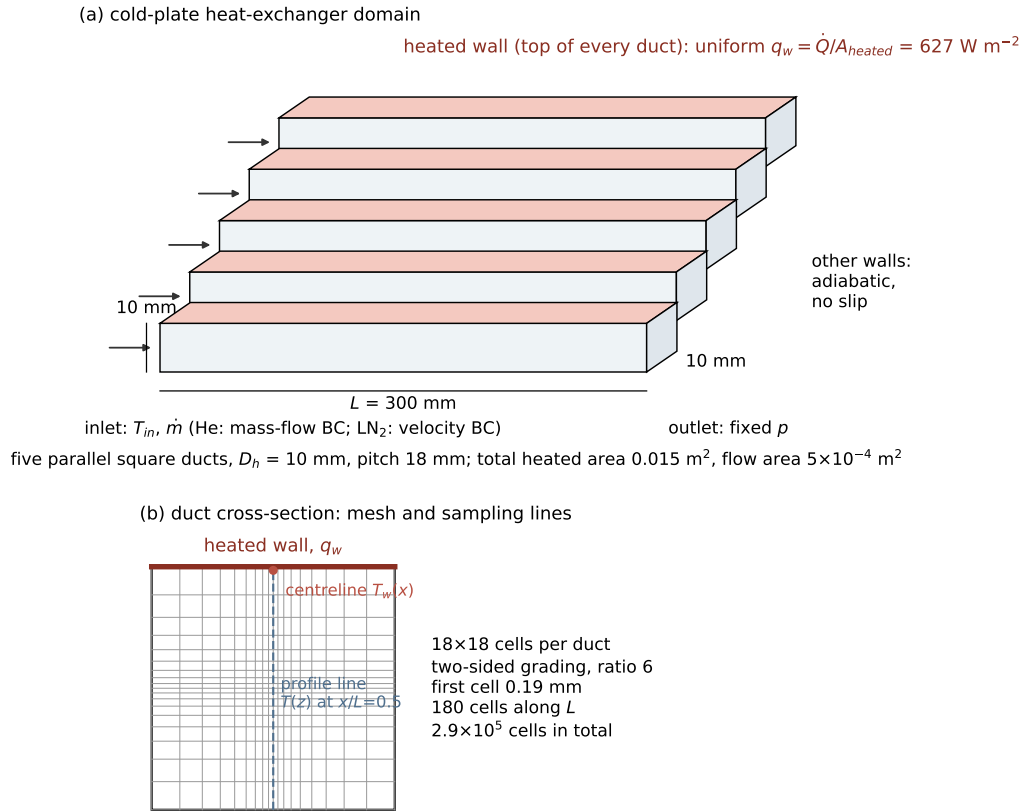


Figure 5: Cold-plate heat-exchanger model: (a) the five heated square ducts with boundary conditions and dimensions; (b) the graded duct cross-section with the two sampling lines used for the wall-temperature profiles.

164 developed square-duct limits $Nu_\infty = 3.61$ and $fRe = 56.9$ [41] and turbu-
 165 lent results against the Gnielinski correlation [42]. Wall temperature profiles
 166 along the heated-wall centreline are extracted for every run.

167 *Grid convergence..* Three systematically refined meshes (refinement ratio
 168 1.5) were run for both fluids at $Re = 2300$ and the discretisation uncertainty
 169 expressed as a GCI [43] (Section 4). For the turbulent runs the first-cell
 170 y^+ on the graded mesh averages 2.5–5.6 with a maximum of 9.5, within the
 171 range handled by the low-Reynolds wall treatment of the SST model.

172 2.4. Work ledger (Q3)

173 A load $\dot{Q}_i$ absorbed at T_i and rejected at $T_0 = 300$ K costs

$$\dot{W}_i = \frac{\dot{Q}_i}{\eta_i} \frac{T_0 - T_i}{T_i}, \quad (7)$$

174 where η_i is the stage efficiency as a fraction of Carnot. The ideal ledger sets
 175 $\eta_i = 1$: the multiplier is 5.0 at 50 K, 2.9 at 77 K and 74 at 4 K. The real ledger
 176 uses published hardware. A commercial two-stage PTR (Cryomech PT415)
 177 delivers 1.5 W at 4.2 K together with 40 W at 45 K for 10.7 kW of input [33];
 178 allocating the input in proportion to the Carnot work of each stage gives
 179 $\eta_2 \approx 0.011$ at 4.2 K and $\eta_1 \approx 0.045$ at 45 K (Supplementary S3), consistent
 180 with the 1 % and 10–15 % of Carnot that Radebaugh reports for 4 K and
 181 80 K coolers [30]. Liquid nitrogen carries 199 kJ kg⁻¹ of latent heat at 1 bar;
 182 its ideal production work is 0.213 kWh kg⁻¹ [2] and real air-separation plants
 183 need 0.5–0.7 kWh kg⁻¹ [31], an exergy efficiency $\eta_{N_2} \approx 0.3$ –0.4. The 4 K stage
 184 of the nitrogen route is still served by a cryocooler, so its η_2 is the same in
 185 both routes. The ranges above are propagated as bands.

186 3. Results Q1: the passive penalty of a 77 K shield

187 3.1. Radiation verification and the view-factor closure

188 The first-stage radiative load computed with the view-factor matrix as
 189 generated does not converge under agglomeration refinement: 9.46, 8.41, 6.49
 190 and 3.26 W at 60, 250, 400 and 800 coarse faces per patch, against 7.07 W
 191 from the parallel-plate closed form (Fig. 6). The cause is closure. In a closed
 192 enclosure every row of the view-factor matrix must sum to one; the generated
 193 rows sum to between 0.50 and 1.13 (mean 0.97–1.01), and finer agglomeration
 194 worsens the scatter. With surface emissivities of 0.02–0.04 the radiosity is

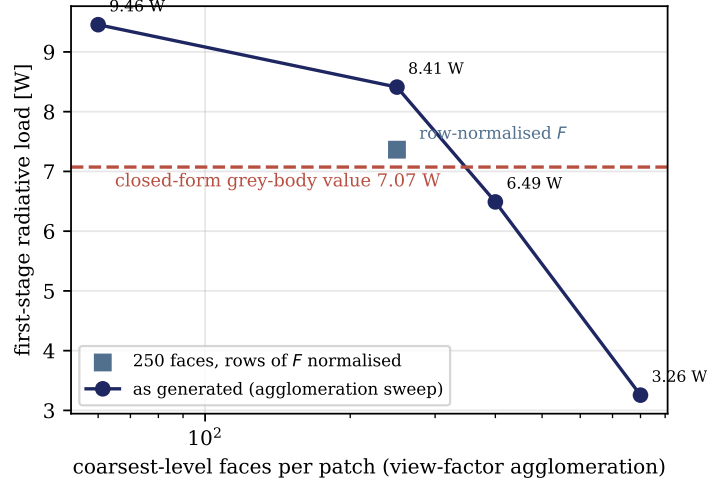


Figure 6: First-stage radiative load ($300 \rightarrow 50$ K) against the view-factor agglomeration level with the matrix as generated (circles) and with its rows normalised to unity at the production level (square), compared with the closed-form grey-body value. The monotonic collapse under refinement is a closure artefact amplified by the low emissivities.

195 reflected of order $1/\varepsilon$ times before it is absorbed, so a per-reflection leak of a
 196 few per cent compounds into a load error of tens of per cent. Normalising each
 197 row of the generated matrix to unity removes the artefact: at the production
 198 level of 250 faces the $300 \rightarrow 50$ K load becomes 7.36 W (4.1 % above the closed
 199 form), the $300 \rightarrow 77$ K load 7.34 W (4.2 %), and the two 4 K-stage loads 3.16
 200 and 17.8 mW (0.5 % above 3.15 and 17.7 mW). The remaining 4 % at the
 201 first stage is the geometric departure of the real enclosure, with its harness
 202 penetrations, from the parallel-plate idealisation. All loads below are the
 203 row-normalised values; the unnormalised production values are retained as
 204 an upper band.

205 3.2. Stage loads and mechanism split

206 At the first stage radiation dominates and the shield temperature barely
 207 matters (Table 3, Fig. 7). The radiative flux scales with $T_{\text{hot}}^4 - T_{\text{cold}}^4$ and 300^4
 208 dwarfs both 50^4 and 77^4 , so the 77 K shield intercepts 99.6 % of what the
 209 50 K shield intercepts. Harness conduction falls with the smaller span, 962
 210 against 1029 mW. The first-stage loads are 8.39 W (50 K) and 8.30 W (77 K);
 211 the emissivity band of ± 50 % moves them between 6.5 and 12.5 W but leaves
 212 the 1 % difference between routes unchanged. The conservative production

Table 3: Stage loads by mechanism (row-normalised view factors; closed-form value in parentheses), with the unnormalised production value and the $\pm 50\%$ emissivity band as upper and lower bounds.

Case	Radiation	Conduction	Total	Prod.	$\varepsilon \pm 50\%$ [W]
300 $\rightarrow$ 50 K	7.36 W (7.07)	1.029 W (1.029)	8.39 W	9.44 W	6.5–12.5
300 $\rightarrow$ 77 K	7.34 W (7.04)	0.962 W (0.962)	8.30 W	9.35 W	6.4–12.4
50 $\rightarrow$ 4 K	3.16 mW (3.15)	70.3 mW (70.3)	73.5 mW	75.6 mW	–
77 $\rightarrow$ 4 K	17.8 mW (17.7)	162.1 mW (162.1)	179.9 mW	191.8 mW	–
300 $\rightarrow$ 50 K MLI	1.81 W ^a (1.47)	1.029 W	2.84 W	–	–
300 $\rightarrow$ 77 K MLI	1.81 W ^a (1.46)	0.962 W	2.77 W	–	–

^aproduction-level matrix; the closure bias of +23% applies as at the bare-shield level.

213 value of 9.4 W is used as the cold-plate duty in Section 4.

214 At the 4 K stage the picture inverts. Residual radiation rises by the factor
215 $(77^4 - 4^4)/(50^4 - 4^4) = 5.6$, but from a base of 3 mW; harness conduction
216 rises by the factor 2.31 from a base of 70 mW. The 77 K route therefore
217 delivers 106 mW more to the 4 K plate: 92 mW of conduction and 15 mW
218 of radiation, 86% conductive. Through the ideal 4 K multiplier of 74 this
219 is 7.9 W of room-temperature power, 6.8 W of it from the harness. Every
220 stage-integrated ratio agrees with its closed form to better than 0.5%.

221 The MLI-equivalent shield ($\varepsilon = 0.003$ on the first-stage surfaces) cuts the
222 first-stage radiation to 1.81 W on either route (closed form 1.47 W with the
223 production-level matrix bias), so the first-stage loads fall to 2.84 and 2.77 W
224 and become conduction-heavy (36% conduction). The 4 K-stage loads are
225 unaffected. This variant is the one that changes the ledger verdict in Sec-
226 tion 5.

227 3.3. Spatial distribution

228 The view-factor solution concentrates the radiative flux on the shield
229 regions with a direct line of sight to the OVC (8.2–10.8 W m⁻² across the
230 shield) and leaves the harness penetrations cooler (Fig. 8). Conduction is
231 shared among the bundles in the ratio of their line counts: on the 77 $\rightarrow$ 4 K
232 span the drive and flux families deliver 125 of the 162 mW. These two bundles
233 are the targets for any intermediate heat sink, and the harness conduction
234 integral, not the cryogen, is the lever on the 4 K-stage penalty.

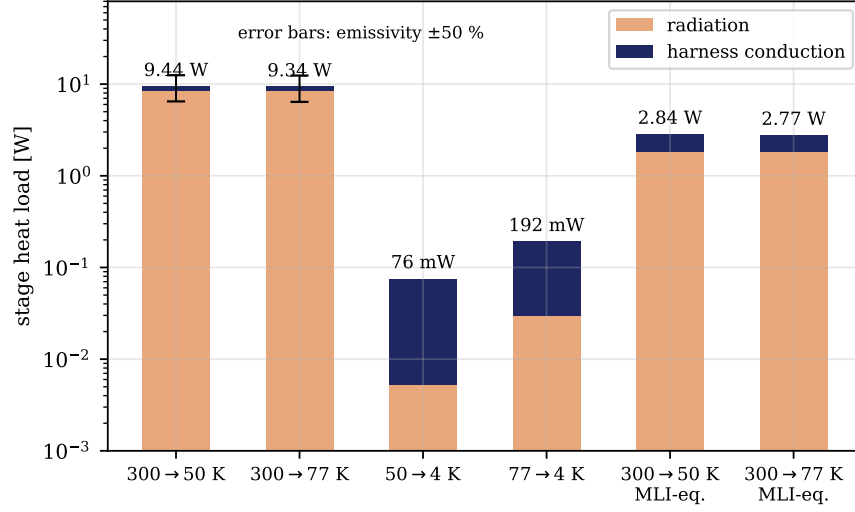


Figure 7: Stage loads for the four enclosure cases and the two MLI-equivalent variants (production view factors), decomposed by mechanism, with the $\pm 50\%$ emissivity band. The first stage is radiation-dominated and nearly identical between routes; the 4 K stage is conduction-dominated and is where the routes diverge.

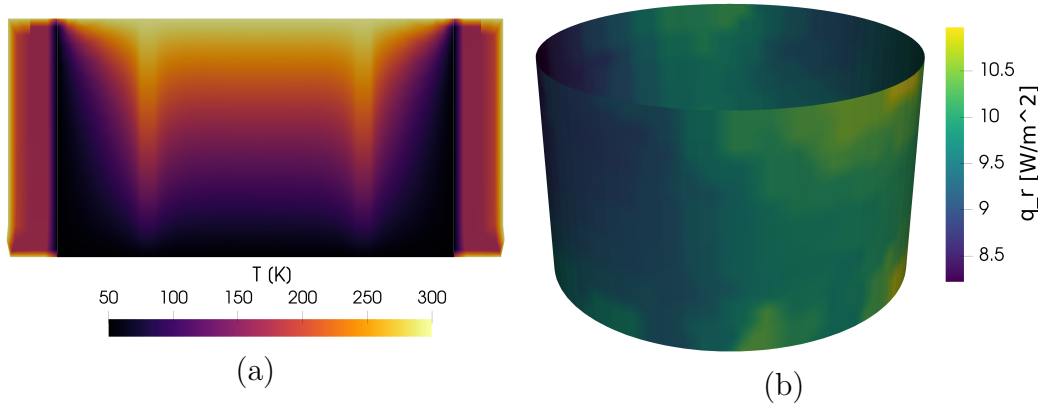


Figure 8: (a) Mid-plane temperature and (b) shield radiative-flux map of the $300 \rightarrow 50$ K case.

Table 4: Grid convergence at $Re = 2300$ (Celik et al. [43]; meshes of 86k, 292k and 984k cells, refinement ratio 1.5). The production mesh is the medium one.

Fluid	Quantity	Coarse	Medium	Fine	p	Extrapolated	GCI_{medium}
He (1 bar)	Nu	6.439	6.449	6.457	0.64	6.49	0.7 %
	f	0.0632	0.0642	0.0647	1.40	0.0655	2.5 %
LN ₂ (3 bar)	Nu	10.51	10.44	10.38	0.87	10.26	2.2 %
	f	0.0592	0.0599	0.0603	1.71	0.0607	1.6 %

235 4. Results Q2: helium gas versus liquid nitrogen in the cold plate

236 4.1. Verification

237 Every forced-convection run closes its energy balance, wall heat input
238 against mass-flux-weighted enthalpy rise, to within 0.2 % (Supplementary
239 Table S7). At $Re = 500$ the helium flow is thermally developed over most
240 of the channel ($L_t/L = 0.59$) and returns $Nu = 3.66$, 1.2 % above the fully
241 developed square-duct value of 3.61 [41]; every other laminar run is ther-
242 mally developing (L_t/L up to 8.8 for nitrogen) and lies above it, as it must.
243 The three-grid studies (Table 4) converge monotonically for both fluids: the
244 production-mesh grid convergence index is 0.7 % (Nu) and 2.5 % (f) for he-
245 lium and 2.2 % and 1.6 % for nitrogen. In the turbulent runs the first-cell y^+
246 on the heated wall averages 2.5–5.6 with a maximum of 9.5, within the low-
247 Reynolds treatment of the SST model. Nitrogen’s turbulent Nusselt numbers
248 lie 8–9 % below the Gnielinski correlation [42] and helium’s 28–32 % below
249 it; the correlation is for a duct heated on all walls, and a single heated wall
250 penalises the low-Prandtl gas, whose thermal layer fills the duct, more than
251 the liquid.

252 4.2. Operating maps

253 Figure 9 maps the two coolants over $Re = 500$ – 10^4 at the fixed 9.4 W
254 duty. Nitrogen’s heat-transfer coefficient exceeds helium’s by a factor of
255 4.9–5.1 throughout the laminar range and 6.2–6.3 in the turbulent range;
256 the laminar ratio is set by the conductivity ratio of 3.1 times the ratio of
257 developing-flow Nusselt numbers, which favours the high-Prandtl liquid. The
258 consequence for the plate is stark. At atmospheric pressure the helium-cooled
259 wall reaches 98 K at $Re = 500$ and 69 K at $Re = 2300$, and falls below the
260 77 K nitrogen temperature only above $Re \approx 1400$ (Fig. 9d); the nominal
261 50 K of the helium route is reached only in the turbulent range, at $Re = 10^4$

and 65 mW of pumping power. The nitrogen-cooled wall stays between 78 and 86 K over the whole map. Its saturation margin at 3 bar is 2.2 K at $Re = 500$, 7.1 K at 2300 and 10.6 K at 10^4 ; at 1 bar the wall would exceed the 77.2 K saturation temperature in every laminar run, so a bath-fed loop at atmospheric pressure would boil on the plate at this duty and a pressurised or subcooled loop is a requirement, not a refinement.

Pumping power separates the fluids in the opposite sense once pressure enters. At 1 bar helium needs 38 times nitrogen's pumping power at matched Reynolds number (1.39 against 0.037 mW at $Re = 2300$), because its mass flow is 26 times smaller but its velocity 32 times larger. Raising the helium pressure at fixed Reynolds number leaves the mass flow, the Nusselt number and the wall temperature unchanged to four significant figures and reduces the pumping power as the inverse square of the density: 25.0 times at 5 bar and 324 times at 18 bar (Fig. 9c). At the 18 bar of the ITER and DTT shield loops [4, 5] helium's pumping power is $4.3 \mu\text{W}$, 8.5 times below nitrogen's; the parallel-path gaseous-helium shields of LHD operate in the same regime [6].

4.3. Wall-temperature distributions

Figure 10 gives the heated-wall centreline temperature and the cross-channel profile at mid-length, the quantitative counterpart of the rendered fields of the earlier comparison. Along the wall the temperature rise grows as $x^{1/3}$ in the developing region and then linearly; at the outlet the helium wall sits 24 K above its inlet at $Re = 2300$ and 53 K at $Re = 500$, nitrogen 4.2 and 8.7 K. Across the channel the helium thermal layer at $Re = 500$ fills the duct, whereas nitrogen's occupies the near-wall 30 % even at $Re = 500$. This is the geometric origin of the heat-transfer ratio and of the plate temperatures in Fig. 9d.

4.4. Ranking on three bases

Table 5 collects the design point and Fig. 11 the trade-off between wall superheat and pumping power. The ranking depends on the basis. At matched Reynolds number nitrogen wins heat transfer by a factor of 5 and pumping power by 38; the effectiveness–NTU pair favours helium (0.11 against 0.06), a capacity-rate artefact of the gas's small $\dot{m}c_p$ that says nothing about how cold the plate is. At matched mass flow nitrogen's Reynolds number falls to about 90 and its heat-transfer coefficient to the fully developed limit of about $53 \text{ W m}^{-2}\text{K}^{-1}$, still 1.8 times helium's at $Re = 2300$. At matched pumping power, the basis a plant designer pays for, nitrogen at the pumping power of

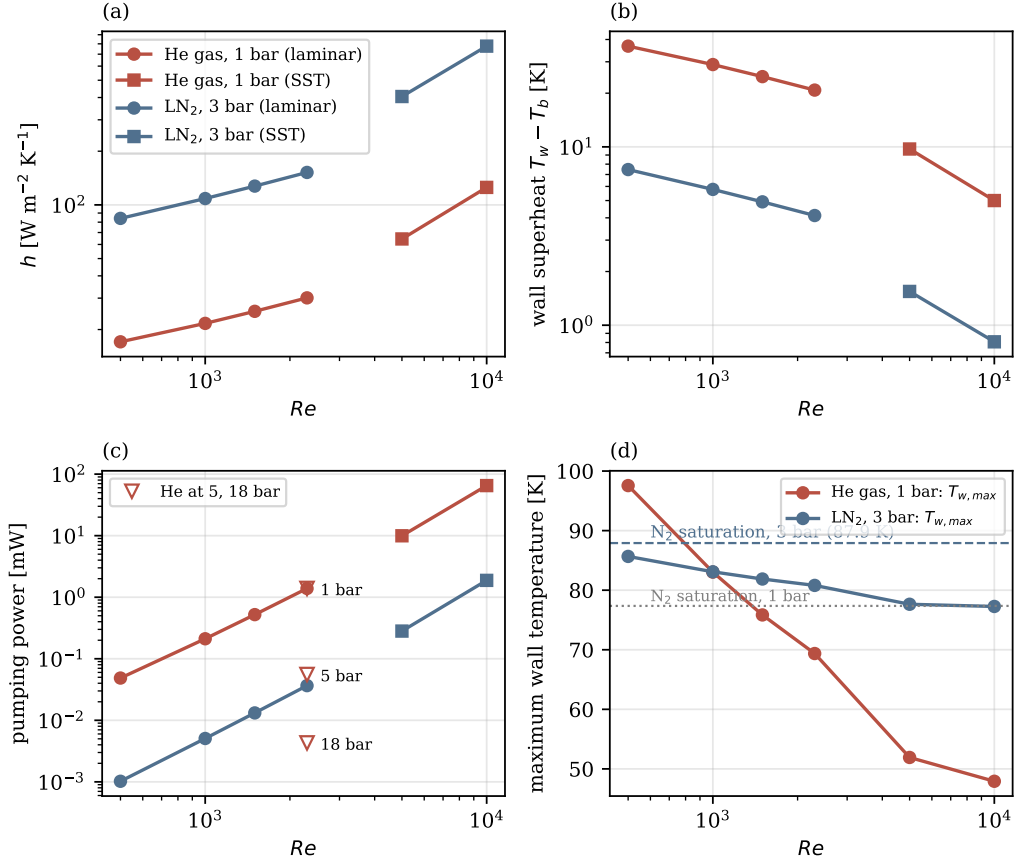


Figure 9: Operating maps at 9.4 W duty: (a) heat-transfer coefficient, (b) wall superheat, (c) pumping power with the helium pressure series at $Re = 2300$, and (d) maximum wall temperature with the nitrogen saturation temperatures at 1 and 3 bar. Circles: laminar; squares: $k-\omega$ SST.

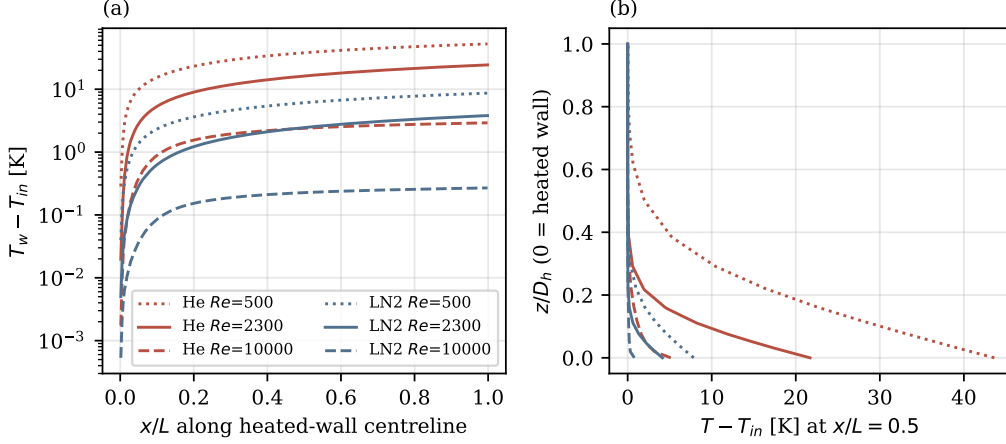


Figure 10: (a) Heated-wall centreline temperature rise above the inlet and (b) cross-channel temperature profile at $x/L = 0.5$ for both fluids at $Re = 500, 2300$ and 10^4 .

298 1 bar helium ($Re = 2300$) holds a superheat of 0.9 K against helium's 20.8 K,
 299 a factor of 23; at the pumping power of 18 bar helium the factor is 3.5 (5.9
 300 against 20.8 K). In the turbulent range the curves in Fig. 11 are parallel: a
 301 given superheat costs helium at 1 bar roughly 30 times the pumping power it
 302 costs nitrogen, and the 18 bar helium point sits inside the nitrogen envelope
 303 on pumping power alone but outside it on superheat. No single number ranks
 304 the coolants; the operating point and the pressure do.

305 4.5. Buoyancy sensitivity

306 The buoyancy parameter Gr/Re^2 of the forced-convection runs is below
 307 0.6 for helium and above unity for nitrogen in the laminar range (40 at
 308 $Re = 500$, 1.05 at 2300), so a real plate in which gravity acts along the heated
 309 wall would carry a mixed-convection secondary flow in nitrogen. The three
 310 runs with gravity on confirm this and bound its effect. Helium converged
 311 and gained 27% in Nu at $Re = 2300$ and 118% at $Re = 500$. Nitrogen did
 312 not reach a steady state at either Reynolds number: after 10 000 iterations
 313 the temperature residual oscillated at 10^{-3} and the energy balance remained
 314 open by 11% and 72%, the signature of an unsteady buoyant flow that a
 315 steady solver cannot represent. The forced-convection nitrogen results are
 316 therefore lower bounds on its heat transfer, and the ranking of Section 4.4 is
 317 conservative toward helium.

Table 5: Design point, $Re = 2300$, 9.4 W, forced convection. He at 1 bar (18 bar in parentheses where pressure matters); LN₂ at 3 bar. GCI on Nu and h : 0.7 % (He), 2.2 % (LN₂).

Quantity	He, 50 K	LN ₂ , 77 K	LN ₂ /He
Mass flow $\dot{m}$ [g s ⁻¹]	0.731	18.8	25.7
Volumetric flow [cm ³ s ⁻¹]	702 (39)	23.3	0.033
Mean velocity U [m s ⁻¹]	1.52 (0.084)	0.047	0.031
Inlet / outlet temperature [K]	45.0 / 47.5	77.0 / 77.2	–
Pressure drop Δp [Pa]	1.98 (0.11)	1.57	0.80
Pumping power $\dot{W}_p$ [μ W]	1388 (4.3)	36.6	0.026 (8.5)
Wall temperature, mean / max [K]	67.1 / 69.4	81.2 / 80.8 ^a	–
Wall superheat $T_w - T_b$ [K]	20.8	4.12	0.20
Heat-transfer coefficient h [W m ⁻² K ⁻¹]	30.1	152	5.05
Nusselt number Nu	6.45	10.4	1.62
Darcy friction factor f	0.064	0.060	0.93
NTU / effectiveness ε	0.119 / 0.112	0.060 / 0.058	0.50 / 0.52
Mouromtseff number	567 (5720)	4.6×10^4	81 (8)
Buoyancy parameter Gr/Re^2	0.02	1.05	–
Saturation margin $T_{\text{sat}} - T_{w,\text{max}}$ [K]	–	7.1	–
Energy-balance closure	0.06 %	0.001 %	–

^acentreline maximum; the area mean includes the corner regions.

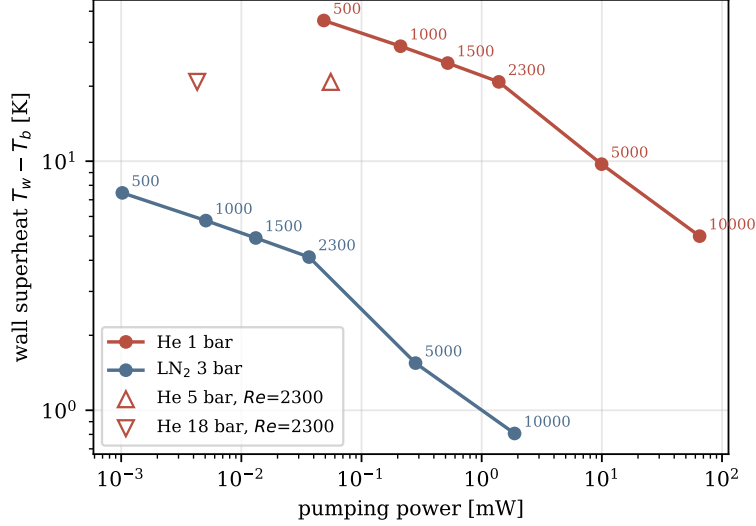


Figure 11: Wall superheat against pumping power for both fluids over the Reynolds sweep (labels: Re), with the helium pressure series at $Re = 2300$.

5. Results Q3: the work ledger

Table 6 and Fig. 12 price the two routes with Eq. (7). With ideal multipliers and the bare shield the nitrogen route is cheaper: 37.3 against 47.4 W, a margin of 21 %, because intercepting 8.3 W at a multiplier of 2.9 instead of 5.0 saves 18 W while the 4 K-stage penalty costs 7.9 W. The emissivity band (31.8–49.2 W against 37.8–68.0 W) does not close the gap. With the MLI-equivalent shield the first-stage load shrinks to 2.8 W, the nitrogen saving to 6.2 W, and the verdict flips: helium 19.6 W against nitrogen 21.3 W. The ideal ledger therefore favours nitrogen where the first-stage load is large and helium where insulation has removed it, which is the regime of the MLI-wrapped shields surveyed by Shu et al. [3] and of the single-layer intercepts of Singh et al. [44].

With hardware efficiencies the verdict is not close in either case. The 4 K stage at 0.8–1.5 % of Carnot turns the 106 mW penalty into 520–980 W, more than the whole first-stage bill of either route, and the helium route costs 840 W (640–1100) against 1290 W (950–1740) for nitrogen with the bare shield and 610 against 1240 W with MLI. Written as a break-even, the

Table 6: Work ledger: stage loads (row-normalised view factors), ideal and hardware-based room-temperature power. Ranges: emissivity $\pm 50\%$ (ideal) and stage-efficiency ranges of Supplementary Table S9 (real).

Shield	Route	$\dot{Q}_1$ [W]	$\dot{Q}_2$ [mW]	$\dot{W}_{\text{ideal}}$ [W]	$\dot{W}_{\text{real}}$ [W]
bare	He, 50 K	8.39	73.5	47.4 (37.8–68.0)	840 (640–1100)
bare	LN ₂ , 77 K	8.30	179.9	37.3 (31.8–49.2)	1290 (950–1740)
MLI-eq.	He, 50 K	2.84	73.5	19.6	610 (460–820)
MLI-eq.	LN ₂ , 77 K	2.77	179.9	21.3	1240 (910–1690)

nitrogen route is cheaper only if

$$\frac{\eta_2}{\eta_1} \gtrsim \frac{74 \Delta \dot{Q}_2}{5.0 \dot{Q}_{1,\text{He}} - 2.9 \dot{Q}_{1,\text{N}_2} \eta_1 / \eta_{\text{N}_2}}, \quad (8)$$

which with the present loads, $\eta_1 = 0.12$ and $\eta_{\text{N}_2} = 0.35$ requires a 4 K-stage efficiency of 2.8 % of Carnot for the bare shield and 8 % for the MLI shield, two to eight times what commercial coolers achieve [30]. The lever is $\Delta \dot{Q}_2$: because 86 % of it is harness conduction, a harness engineered for a threefold lower $77 \rightarrow 4$ K conduction integral, by heat-sinking the drive and flux bundles or by lower-conductivity outer conductors, brings the bare-shield break-even to 1.2 % of Carnot, within the range of real hardware.

6. Discussion

The penalty is a harness property.. The passive cost of the warmer shield is located, sized and attributed: 106 mW at the 4 K stage, 86 % of it conduction along a harness whose drive and flux families carry three quarters. It is a property of the wiring, not of the cryogen, and Eq. (8) shows that the wiring is the only variable that can move the break-even into the range of real hardware. Lumped budgets give the same total; what the resolved model adds is the attribution to bundles that can be heat-sunk, and the confirmation that the residual radiation, whatever the shield temperature, is a minor term. This is the design steer that Li et al.’s optimisation of intercepts on supports and in insulation [11] cannot give for a wired cryostat.

The helium plate floats.. A cold plate cooled by circulated helium gas at atmospheric pressure does not sit at the gas temperature. At the 9.4 W duty it runs 21 K above the bulk at $Re = 2300$ and reaches nitrogen temperature

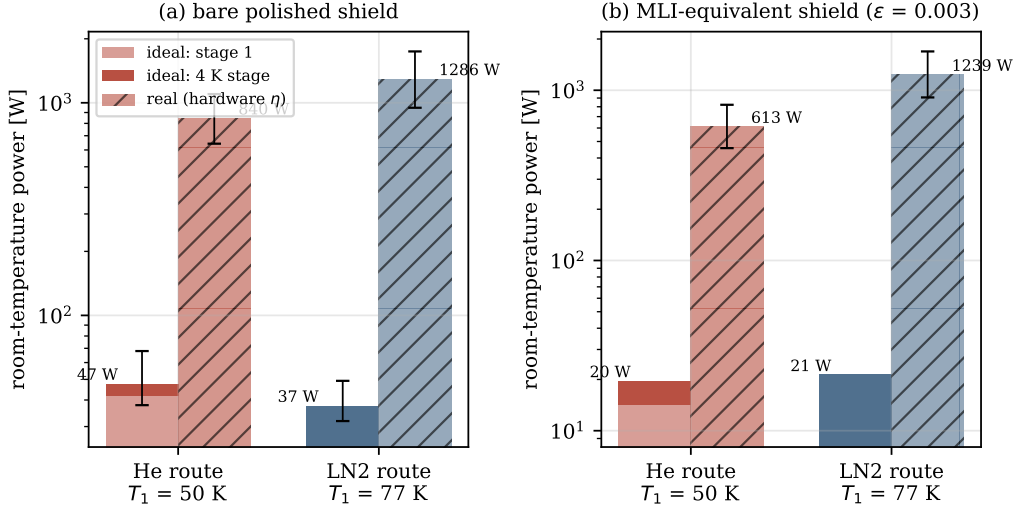


Figure 12: Room-temperature power of the two routes for (a) the bare polished shield and (b) the MLI-equivalent shield: ideal Carnot multipliers split by stage (solid) and hardware efficiencies (hatched), with emissivity and efficiency ranges as error bars.

at $Re \approx 1400$; the nominal 50 K is recovered only in turbulent flow at tens of milliwatts of pumping power, or by pressurising the loop, which leaves the wall temperature untouched but makes the pumping power negligible. This is a statement about the circulated-gas architecture and the channel sizing, not about the conductively coupled PTR first stage of Fig. 2a, whose plate temperature is set by the cooler's capacity curve. It does mean that the 50/77 K distinction of the passive comparison is only realised on the helium route with an adequate flow or pressure, and Section 4.2 gives the conditions.

The basis of comparison decides the ranking.. Matched Reynolds number, matched mass flow and matched pumping power rank the fluids differently in magnitude and, for pumping power at 18 bar, in sign. The effectiveness–NTU pair inverts the engineering conclusion because it rewards a small capacity rate. A cryogenic coolant comparison must therefore state its basis and its operating point, as Gačnik et al. found necessary for nitrogen-cooled terminals [29]; the operating maps of Fig. 9 are offered as the form such a comparison should take.

Ideal and real ledgers disagree, and the disagreement is informative.. The ideal ledger prefers nitrogen for a bare shield and helium for an insulated

one; the hardware ledger prefers helium in both cases by 35–50 %, because the 4 K stage of a commercial cooler converts every milliwatt of penalty into ten watts at the wall. The nitrogen route retains what Li et al. identified three decades ago [11]: a constant shield temperature, stored reserve, and independence from the cryocooler, which now also means freedom from its vibration [45]. Those are operational arguments; the thermodynamic one is conditional on the harness.

Methodological findings. Two travel beyond this geometry. First, view-factor matrices generated for low-emissivity enclosures must be checked for closure and, if the enclosure is closed, row-normalised; the required closure accuracy is of order the emissivity, and agglomeration refinement does not supply it. The benchmark practice of Končar et al. [25] should be extended by this test. Second, the mixing-cup temperature on a graded mesh must be mass-flux weighted without an additional area factor; the corrected accounting closed the energy balance of every run from a few per cent to below 0.2 %.

Limitations. The enclosure imposes fixed temperatures on its plates and shield and neglects contact resistance at the harness ends; a finitely conductive shield would carry the radiative-flux map of Fig. 8 into a temperature map, which is the next step. The cold plate is a single-wall-heated smooth duct with constant properties except for the helium density and the nitrogen buoyancy term; conjugate conduction in the plate, which would spread the wall temperature, is not resolved. Nitrogen is single phase by verified margin at 3 bar but not at 1 bar. The turbulent runs rely on the SST model at $y^+ \leq 9.5$ and were not compared with experiment. The hardware efficiencies are ranges from published data, and the allocation of a two-stage cooler’s input between its stages is by Carnot work; a measured capacity map of a specific machine would replace the ranges. Buoyancy in nitrogen was shown to be unsteady and was excluded from the ranking rather than resolved; a transient calculation in a specified plate orientation would quantify it.

7. Conclusions

1. A 77 K first shield costs nothing at the first stage relative to a 50 K shield: it intercepts 99.6 % of the radiation and 6.5 % less harness conduction, 8.3 against 8.4 W for the enclosure studied. The penalty appears at the 4 K stage, 106 mW, and is 86 % harness conduction

- concentrated in the drive and flux bundles; it is worth 7.9 W of ideal room-temperature power and 520–980 W with real 4 K coolers.
2. View-factor radiation in low-emissivity enclosures is controlled by matrix closure: the generated matrix gave loads from 9.5 to 3.3 W under agglomeration refinement against a 7.07 W closed form, and row normalisation recovered the closed form to 4 % at the first stage and 0.5 % at the 4 K stage.
 3. As a first-shield coolant at 9.4 W, nitrogen at 3 bar gives 5 times helium’s heat-transfer coefficient in laminar flow and 6 times in turbulent flow, and holds the plate within 4 K of its inlet at $Re = 2300$ against helium’s 21 K; helium at 1 bar lets the plate float above 77 K below $Re \approx 1400$ and needs 38 times the pumping power, but at 18 bar its pumping power falls below nitrogen’s by a factor of 8.5 with heat transfer unchanged. The ranking depends on the basis of comparison and the operating point.
 4. The ideal ledger favours nitrogen by 21 % for a bare shield and helium by 8 % for an MLI shield; with hardware efficiencies helium is cheaper by 35–50 % in both cases. The nitrogen route becomes competitive with real hardware only if the 77 → 4 K harness conduction is cut by about a factor of three.
 5. The corrected helium properties, the forced-convection basis, the mass-flux-weighted mixing-cup temperature and the row-normalised view factors together change the earlier quantitative picture of this comparison; the qualitative conclusion, that the shield choice and the harness design are not separable, is unchanged and now rests on verified numbers.

CRediT authorship contribution statement

Imerson Joao: Conceptualization, Methodology, Software, Investigation, Formal analysis, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

443 **Data availability**

444 All OpenFOAM cases (enclosure baseline and variants, cold-plate sweep,
445 grid studies), the post-processing and figure scripts, and the tabulated results
446 are available at [github.com/Imerson/Pre-cooling-a-superconducting-qubit-dilution-](https://github.com/Imerson/Pre-cooling-a-superconducting-qubit-dilution-refrigerator)
447 refrigerator (release v2); an archived copy will be deposited on Zenodo on
448 acceptance.

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452 **Declaration of generative AI in scientific writing**

453 During the preparation of this work the author used Claude (Anthropic)
454 to assist with editing, revision and consistency checking of the manuscript
455 and with scripting the simulation campaign. The author reviewed and edited
456 all content and takes full responsibility for the published article.

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